SwarmPerf Theme Demo

A fast, minimal showcase of the Performance Theme

LaTeX Helper Swarm Project

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1 Introduction

This document demonstrates the `swarmperf` theme, a minimal, fast-to-compile LaTeX document style designed for speed and simplicity. Unlike the full-featured `swarmbeauty` theme, `swarmperf` deliberately excludes heavy packages such as TikZ, fontspec, microtype, and tcolorbox to minimize compilation time and PDF output size.

The design philosophy is simple: every package must earn its place. The result is a theme that compiles **2–5x faster** than typical modern LaTeX setups while still producing clean, professional documents suitable for technical reports, academic papers, and internal documentation.

1.1 Motivation

Many LaTeX documents do not need elaborate decorations, syntax-highlighted code blocks via external tools, or custom OpenType fonts. For these use cases, the overhead of loading fontspec, minted, tcolorbox, and TikZ can add significant compilation time—especially on CI/CD servers or in iterative edit-compile cycles.

The `swarmperf` theme provides sensible defaults for clean document formatting without the weight.

Note

This theme works with **pdfLaTeX**, **XeLaTeX**, and **LuaLaTeX**. No `--shell-escape` flag is required.

2 Typography and Text

The theme uses the default Computer Modern fonts (or Latin Modern if available). Text is set with sans-serif headings and serif body text. Mathematics uses the standard math fonts.

2.1 Inline Elements

You can highlight **important terms** using the `hltext` command. For inline code, use `code`: `some_function()`. For code with special characters: `text_with_underscores`.

2.2 Lists

- First item in an unordered list
- Second item with **emphasis**
- Third item with nested content:
  - Nested item A
  - Nested item B

1. Step one: load the performance theme
2. Step two: write your content
3. Step three: compile with any LaTeX engine

3 Block Environments

The theme provides lightweight block environments for notes, tips, and warnings. These use basic minipage and color commands—no tcolorbox overhead.

Note

This is a **note** block. Use it for general observations, additional context, or supplementary information that is relevant but not critical to the main argument.

Tip

This is a **tip** block. Use it to highlight best practices, shortcuts, or recommendations that can improve the reader's workflow or understanding.

Warning

This is a **warning** block. Use it to alert the reader about potential pitfalls, common mistakes, or deprecated features that they should be aware of.

4 Tables

The theme integrates with `booktabs` for clean, professional tables.

4.1 Booktabs Table

Table 1 Comparison of LaTeX compilation engines

Engine	Speed	Font Support	Unicode
pdfLaTeX	Fast	Limited (Type 1)	Partial
XeLaTeX	Medium	Full (OpenType)	Full
LuaLaTeX	Slow	Full (OpenType)	Full

5 Code Listings

The `swarmperf` theme uses the `listings` package for code highlighting. Unlike `minted`, `listings` requires no external dependencies (no Pygments, no `--shell-escape`).

```
def fibonacci(n: int) -> list[int]:
    """Generate the first n Fibonacci numbers."""
    seq = [0, 1]
    for i in range(2, n):
        seq.append(seq[-1] + seq[-2])
    return seq[:n]

if __name__ == "__main__":
    result = fibonacci(10)
    print(f"First 10: {result}")
```

Fibonacci sequence generator

```
\begin{equation}
  E = mc^2
  \label{eq:einstein}
\end{equation}
```

A simple equation

6 Mathematics

The theme provides standard theorem environments with automatic numbering.

Theorem 6.1 (Euler's Identity) *For any real number x, the following identity holds:*

$$e^{i\pi} + 1 = 0$$

Definition 6.2 (Metric Space) *A* metric space *is a pair (X, d) where X is a set and $d\colon X \times X \to \mathbb{R}_{\geq 0}$ satisfies:*

1. $d(x, y) = 0 \iff x = y$ *(identity of indiscernibles)*
2. $d(x, y) = d(y, x)$ *(symmetry)*
3. $d(x, z) \leq d(x, y) + d(y, z)$ *(triangle inequality)*

Lemma 6.3 (Nested Intervals) *Let $\{[a_n, b_n]\}_{n=1}^{\infty}$ be a sequence of closed, bounded intervals such that $[a_{n+1}, b_{n+1}] \subseteq [a_n, b_n]$ for all n. Then $\bigcap_{n=1}^{\infty}[a_n, b_n] \neq \emptyset$.*

6.1 Inline and Display Math

The quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
A more complex display equation:

$$\int_{-\infty}^{\infty} e^{-x^2}\, dx = \sqrt{\pi}$$

7 Summary

This demo has showcased the following features of the `swarmperf` theme:

1. **Title page** with colored accent rule
2. **Section headings** with clean sans-serif formatting
3. **Block environments** (note, tip, warning)
4. **Styled tables** with booktabs
5. **Syntax-highlighted code** via listings (no external deps)
6. **Theorem environments** with section numbering
7. **Inline code and emphasis** commands
8. **Headers and footers** with page numbers

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